GPO OPERATIONAL HANDBOOK

GPO Essentials, Processing Order & Technical Implementation

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SECTION 1

GPO Essentials

Understanding Group Policy Objects — the foundation of Windows environment management

1.1 What Is Group Policy (GPO)?

Group Policy is a feature of Microsoft Windows that enables centralized management and configuration of operating systems, applications, and user settings in an Active Directory environment. A Group Policy Object (GPO) is a virtual collection of policy settings that is linked to Active Directory containers — Sites, Domains, or Organizational Units (OUs) — and applied to the computers and users within those containers.

GPOs allow IT administrators to enforce security settings, deploy software, redirect folders, manage desktop environments, and control hundreds of system behaviors — all without touching individual machines.

1.2 Core GPO Components

Component	Description
GPO (Group Policy Object)	Container holding policy settings; linked to AD Sites, Domains, or OUs.
GPMC	Group Policy Management Console — the primary GUI tool for managing GPOs.
SYSVOL	Shared folder on domain controllers storing GPO templates and scripts.
ADM / ADMX Templates	Administrative templates defining configurable policy settings.
Computer Configuration	Settings applied to the machine regardless of who logs in.
User Configuration	Settings applied to the user profile regardless of which machine is used.
OU (Organizational Unit)	AD container to which GPOs are linked; represents departments, roles, or locations.
WMI Filter	Query that conditions GPO application on hardware/OS criteria.
Security Filtering	Restricts GPO application to specific users or computer groups.
Loopback Processing	Applies User Configuration settings based on the computer's OU, not the user's.

1.3 GPO Processing Order (LSDOU)

GPOs are processed in the following order. Later policies override earlier ones unless Block Inheritance or Enforced flags are set:

- **L — Local Policy:** Applied first; settings defined directly on the local computer.
- **S — Site:** GPOs linked to the Active Directory site.
- **D — Domain:** GPOs linked at the domain level.
- **OU — Organizational Unit:** GPOs linked to the OU containing the object, from parent OU to child OU.

> **NOTE:** Within each level, multiple GPOs are processed in reverse link order — the GPO highest in the link order is processed last and therefore wins on conflicting settings.

1.4 GPO Value in Enterprise IT

- Eliminates manual per-machine configuration — enforce thousands of settings from a single console.
- Provides consistent security baseline across all endpoints.
- Enables rapid deployment of software, scripts, and configurations.
- Reduces support overhead through standardized user environments.
- Supports compliance and auditing requirements through enforced policy settings.

SECTION 2

GPO Processing Order

From policy creation to client application — the end-to-end workflow

The GPO processing order defines the sequential workflow from policy creation through final enforcement on client endpoints. Adherence to this order ensures consistent application, audit readiness, and predictable behavior.

2.1 Processing Order Overview

#	Stage	Description	Owner
1	**Policy Requirement**	Business or security requirement identified and documented	Business / Security Team
2	**GPO Design**	Settings scoped, OU target identified, naming defined	IT / GPO Admin
3	**Change Approval**	Change request raised and approved via CAB or change process	Change Manager
4	**Test Environment**	GPO created and tested in lab/test OU before production	GPO Admin
5	**GPO Creation**	GPO created in GPMC with correct settings and naming	GPO Admin
6	**Security Filtering**	Apply/Remove Authenticated Users; add target security group	GPO Admin
7	**OU Linking**	Link GPO to target OU; set link order appropriately	GPO Admin
8	**Client Refresh**	Policy propagates at next refresh cycle or forced via gpupdate	Automatic / Admin
9	**Verification**	Confirm application via RSOP, gpresult, or Event Viewer	GPO Admin
10	**Documentation**	Update GPO register with settings, justification, and approver	GPO Admin

2.2 GPO Refresh Intervals

Group Policy is not applied instantaneously. Understanding refresh intervals prevents unnecessary escalations:

Scenario	Default Interval	Notes
Domain computers (background refresh)	Every 90 min ± 30 min offset	Prevents simultaneous DC overload
Domain Controllers	Every 5 minutes	More frequent due to security role
Forced refresh	Immediate	gpupdate /force at command prompt
Computer startup	On boot	Computer Configuration applied before logon
User logon	At logon	User Configuration applied at each logon

SECTION 3

Real-World Implementation Scenarios

Practical GPO deployments with step-by-step guidance

3.1 Scenario: New Department Onboarding

Business Context: The Finance department is being added to the domain. 80 new users and 80 workstations must receive department-specific settings including mapped drives, printer assignments, restricted desktop, and password policy.

Implementation Steps

Step 1: Create OU Structure — In ADUC, create OU=Finance under OU=Departments. Create sub-OUs for Computers and Users.

Step 2: Create Security Group — Create AD group GPO_Finance_Users and add all Finance user accounts.

Step 3: Create Drive Map GPO — New GPO: 'DEPT-Finance-DriveMaps'. Under User Config > Preferences > Drive Maps, map \\fileserver\Finance to F:.

Step 4: Create Printer GPO — New GPO: 'DEPT-Finance-Printers'. Under User Config > Preferences > Printers, add shared printer \\printserver\FinancePrinter01.

Step 5: Apply Security Filtering — Remove Authenticated Users. Add GPO_Finance_Users to security filtering on both GPOs.

Step 6: Link GPOs — Link both GPOs to OU=Finance\Users.

Step 7: Test Pilot — Log in as a test Finance user on a test machine. Run gpupdate /force. Verify F: drive and printer appear.

Step 8: Validate & Document — Run gpresult /r on the test machine. Confirm both GPOs appear in Applied GPOs. Log in the GPO register.

> **TIP:** Use Item-Level Targeting within Drive Map preferences to apply settings only when the user is a member of a specific group — adding an extra layer of precision without additional GPOs.

3.2 Scenario: Security Lockdown Rollout

Business Context: Following a security audit, IT must enforce USB blocking, disable AutoRun, enforce screen lock after 10 minutes, disable Command Prompt for standard users, and enable Windows Firewall on all workstations.

Implementation Steps

Step 1: Create Baseline Security GPO — New GPO: 'SEC-Baseline-Workstations'. Link to OU=Workstations.

Step 2: USB Blocking — Computer Config > Admin Templates > System > Removable Storage Access. Set 'All Removable Storage classes: Deny all access' to Enabled.

Step 3: Disable AutoRun — Computer Config > Admin Templates > Windows Components > AutoPlay Policies. Set 'Turn off AutoPlay' to Enabled for All Drives.

Step 4: Screen Lock — User Config > Admin Templates > Control Panel > Personalization. Enable 'Enable screen saver', set timeout to 600 seconds, enable 'Password protect the screen saver'.

Step 5: Disable CMD — User Config > Admin Templates > System. Set 'Prevent access to the command prompt' to Enabled.

Step 6: Enable Firewall — Computer Config > Windows Settings > Security Settings > Windows Firewall. Set Domain Profile to On.

Step 7: Test on Pilot Group — Create test OU, move 5 pilot machines. Link GPO. Reboot and verify each setting.

Step 8: Phased Rollout — After pilot sign-off, move remaining workstations OU objects. Monitor Event Viewer for policy errors.

> **WARNING:** Never apply a restrictive security GPO directly to the Domain level without testing. A misconfigured GPO at domain level can lock out all users including administrators. Always test in an isolated OU first.

3.3 Scenario: Remote Work Policy Deployment

Business Context: The organization moves to hybrid working. Remote laptops need VPN-enforced DNS settings, BitLocker encryption, disabled local admin rights, and redirected Desktop/Documents folders to OneDrive sync paths.

Implementation Steps

Step 1: Create Remote Laptops OU — Create OU=RemoteLaptops under OU=Computers. Move all laptop objects.

Step 2: BitLocker GPO — New GPO: 'SEC-BitLocker-Laptops'. Computer Config > Admin Templates > Windows Components > BitLocker. Enable 'Require additional authentication at startup'. Store recovery keys in AD.

Step 3: Remove Local Admin — Computer Config > Windows Settings > Security Settings > Restricted Groups. Add BUILTIN\Administrators with only Domain Admins as members.

Step 4: Folder Redirection — New GPO: 'USR-FolderRedirect-Remote'. Redirect Desktop and Documents to \\fileserver\users\%username% or OneDrive local sync path.

Step 5: DNS Settings — Computer Config > Admin Templates > Network > DNS Client. Set DNS suffix and preferred DNS servers to internal DNS.

Step 6: Link & Filter — Link all GPOs to OU=RemoteLaptops. Security filter to GPO_RemoteWorkers group.

Step 7: User Communication — Notify users of folder redirection change — first logon after policy applies will redirect existing data.

NOTE: BitLocker GPO requires the TPM chip to be enabled in BIOS/UEFI on the laptop. Test on a hardware-representative device before mass deployment.

3.4 Scenario: Software Standardization Drive

Business Context: The organization standardizes on Microsoft 365 Apps, Adobe Acrobat Reader, and a custom internal tool. All must be deployed silently to domain-joined machines without user intervention.

Step 1: Package Software — Obtain or create MSI/MST packages for each application. Place in \\fileserver\GPOSoftware\ with appropriate share/NTFS permissions.

Step 2: Create Software Deployment GPO — New GPO: 'APP-StandardSoftware-All'. Computer Config > Software Settings > Software Installation.

Step 3: Add Packages — Right-click Software Installation > New > Package. Browse to UNC path of each MSI. Select Advanced for MST transforms.

Step 4: Set Deployment Type — Set to Assigned (Computer) so software installs silently on next reboot regardless of logged-in user.

Step 5: Link to Target OU — Link to OU=Workstations. Machines will receive software on next reboot after policy refresh.

Step 6: Verify Installation — Check Add/Remove Programs on a test machine post-reboot. Review Application Event Logs for MSI installer entries.

TIP: Use the 'Maintenance' settings in GPO Software Installation to deploy updates (MSP patches) to previously deployed software without requiring a new package.

SECTION 4

USB Blocking & Folder Redirection

Securing removable media and centralizing user data storage

4.1 USB Device Blocking via GPO

USB blocking prevents unauthorized data exfiltration and malware introduction via removable storage devices. GPO provides two complementary approaches: blocking removable storage access and disabling device installation.

Method A: Removable Storage Access Restriction

Path: Computer Configuration > Administrative Templates > System > Removable Storage Access

Policy Setting	Value	Effect
All Removable Storage classes: Deny all access	**Enabled**	Blocks all USB storage read/write
Removable Disks: Deny read access	**Enabled**	Prevents reading from USB drives
Removable Disks: Deny write access	**Enabled**	Prevents writing to USB drives
CD and DVD: Deny all access	**Enabled**	Blocks optical drives
Custom Classes: Deny read access	**Enabled**	Blocks other removable media

Method B: Device Installation Restriction

Path: Computer Configuration > Administrative Templates > System > Device Installation > Device Installation Restrictions

- Enable 'Prevent installation of removable devices' to block any new USB device installation.
- Enable 'Prevent installation of devices not described by other policy settings' for a whitelist-only approach.
- Use 'Allow installation of devices that match any of these device IDs' to whitelist approved peripherals (keyboards, mice) by Hardware ID.

Whitelisting Approved USB Devices

To allow specific approved USB devices (e.g., hardware tokens, approved encrypted drives) while blocking others:

1. Open Device Manager on a machine with the approved device connected.
2. Right-click the device > Properties > Details tab.
3. Select 'Hardware Ids' from the Property dropdown. Copy the full Hardware ID string.
4. In GPO: Computer Config > Device Installation Restrictions > Allow installation of devices that match any of these device IDs.
5. Add the Hardware ID. Ensure this policy is linked and processed before the blocking policy.

```
# Example: Get Hardware ID of connected USB device via PowerShell
Get-PnpDevice -Class USB | Select-Object FriendlyName, InstanceId, Status


# Get specific Hardware IDs for a device
Get-PnpDeviceProperty -InstanceId 'USB\VID_0781&PID_5567\...' -KeyName
DEVPKEY_Device_HardwareIds
```

WARNING: USB blocking affects ALL removable storage including hardware security keys (YubiKey). Identify and whitelist all business-critical USB devices BEFORE enforcing the block policy.

4.2 Folder Redirection Configuration

Folder Redirection moves Windows special folders (Desktop, Documents, Downloads, AppData) from the local user profile to a network location — ensuring data is backed up, roaming-ready, and centrally managed.

Pre-Requisites

- A file server share accessible to all users: e.g., \\fileserver\UserData$
- Share permissions: Full Control to Creator Owner and Domain Admins only.
- NTFS permissions: Each user should have access only to their own folder (use %username% variable).
- The target network path must be available before folder redirection policy applies, or users will receive an error.

GPO Configuration Steps

Path: User Configuration > Policies > Windows Settings > Folder Redirection

Step 1: Open Folder Redirection — In GPMC, navigate to the target GPO > User Config > Windows Settings > Folder Redirection.

Step 2: Right-click target folder — e.g., Documents > Properties.

Step 3: Set Setting — Choose 'Basic – Redirect everyone's folder to the same location'.

Step 4: Set Target Root Path — Enter \\fileserver\UserData$\%username%\Documents (the %username% variable auto-creates per-user subfolders).

Step 5: Configure Options tab — Enable 'Grant the user exclusive rights to Documents'. Enable 'Move the contents of Documents to the new location'.

Step 6: Repeat for other folders — Apply the same process to Desktop, Downloads, and AppData\Roaming as required.

Step 7: Link GPO — Link to the target Users OU. Policy applies at next logon.

Folder	Recommended Target Path	Notes
Documents	\\srv\users$\%username%\Documents	Most critical — contains user work files
Desktop	\\srv\users$\%username%\Desktop	Roams desktop items across machines
Downloads	\\srv\users$\%username%\Downloads	Prevents local disk fill; allows admin cleanup
AppData\Roaming	\\srv\users$\%username%\AppData	Careful — can slow login; evaluate app-by-app
Start Menu	\\srv\users$\%username%\Start Menu	Rarely needed in Windows 10/11 environments

NOTE: AppData\Roaming redirection can significantly increase logon times if the folder is large. Consider using Roaming User Profiles or Citrix Profile Management instead of GPO folder redirection for AppData in VDI/terminal server environments.

4.3 Testing & Validation

Before production rollout, validate USB blocking and folder redirection in a controlled test:

USB Blocking Test

6. Log onto a test machine as a standard user.
7. Run gpupdate /force and reboot.
8. Insert a USB flash drive. Windows should show 'This operation has been cancelled due to restrictions'.
9. Test that approved whitelisted devices (keyboard, mouse, hardware token) still function.
10. Review Event ID 6416 in the Security event log — device installation attempts are logged.

Folder Redirection Test

11. Log onto a test machine as the test user.
12. Open Documents — the path shown in Properties should be the network UNC path.
13. Create a test file. Verify it appears on the file server at \\fileserver\UserData$\testuser\.

14. Log onto a second test machine as the same user. Confirm the file is accessible — verifying roaming behavior.
15. Confirm no data was lost from the user's original local Documents folder.

SECTION 5

MSI Software Deployment

Silent, policy-driven software distribution via Group Policy

5.1 Prerequisites & Packaging

GPO-based software deployment uses Windows Installer (MSI) packages delivered from a network share. Before configuring the GPO, the following must be in place:

- **MSI Package:** Obtain the vendor's MSI file or repackage the application using tools like Advanced Installer, Repackager, or ORCA.
- **Transform File (.MST):** Optional. Used to customize installation settings (serial keys, install path, feature selection) without modifying the original MSI.
- **Network Share (UNC Path):** Create a shared folder accessible to all target computers — e.g., \\fileserver\GPOSoftware\AppName\. Use a dollar-sign share (GPOSoftware$) to hide it from casual browsing.
- **Share Permissions:** Domain Computers (or the specific security group) must have at minimum Read access to the share and NTFS folder.
- **Test MSI silently first:** Run msiexec /i "\\server\share\app.msi" /qn /l*v C:\Logs\install.log on a test machine to confirm silent install works.

```
# Silent install test before GPO deployment
msiexec /i "\\fileserver\GPOSoftware$\AdobeReader\AcroRead.msi" /qn /l*v
C:\Logs\AdobeInstall.log


# Silent install with transform file
msiexec /i "\\fileserver\GPOSoftware$\App\app.msi"
TRANSFORMS="\\fileserver\GPOSoftware$\App\custom.mst" /qn


# Check exit code (0 = success, 3010 = reboot required)
echo Exit code: %errorlevel%
```

5.2 GPO-Based MSI Deployment Steps

Step 1: Open GPMC — Launch Group Policy Management Console. Navigate to the target OU.

Step 2: Create or Edit GPO — Create new GPO: e.g., 'APP-AdobeReader-Workstations'. Right-click > Edit.

Step 3: Navigate to Software Installation — Computer Configuration > Policies > Software Settings > Software Installation (for machine-wide install).

Step 4: Add New Package — Right-click Software Installation > New > Package. In the Open dialog, type the full UNC path (not mapped drive) and select the MSI file.

Step 5: Choose Deployment Method — Select 'Advanced' to configure properties before deploying. Never use a local drive path — it must be a UNC path accessible from all target computers.

Step 6: Configure Deployment Tab — Set Deployment type: Assigned. Uncheck 'Uninstall this application when it falls out of the scope of management' only if desired.

Step 7: Add Transform (if needed) — Click Modifications tab > Add > Browse to the .MST file on the same UNC share.

Step 8: Close & Link — Close Group Policy Editor. Link the GPO to the target OU. Software will deploy on machines' next reboot after policy refresh.

> **WARNING:** Always use UNC paths (\\server\share\file.msi) — NEVER mapped drive letters. GPO software deployment runs under the SYSTEM account, which does not have access to user-mapped drives.

5.3 Assigned vs. Published Deployment

Attribute	Assigned	Published
Available under	Computer Config & User Config	User Config only
Installation trigger	Automatic at startup (Computer) or logon (User)	Manual via Add/Remove Programs or document activation
User choice	No — installs regardless of user preference	Yes — user chooses to install
Auto-repair	Yes — reinstalls if removed or corrupted	No
Best used for	Mandatory tools (security agents, standard apps)	Optional apps users may optionally install
Reboot required	Typically yes (Computer-Assigned)	No (installs on-demand)

5.4 Updating & Removing Deployed Software

Deploying an Update (New MSI Version)

- In the Software Installation node, right-click the existing package > All Tasks > Redeploy application (forces reinstall) or select Upgrades to link the new version to the old.
- For patch-only updates (.MSP files): right-click the existing package > Properties > Upgrades tab > Add the new package as an upgrade.
- For major version upgrades, create a new package entry, go to Upgrades tab, and specify it replaces the previous version.

Removing Deployed Software

- Right-click the software package in GPMC > All Tasks > Remove.

- Choose 'Immediately uninstall the software from users and computers' for forced removal at next policy refresh/reboot.
- Choose 'Allow users to continue to use the software, but prevent new installations' for a softer retirement.

```
# Verify installed software on a target machine
Get-WmiObject -Class Win32_Product | Select-Object Name, Version | Sort-Object Name


# Check GPO software installation events
Get-EventLog -LogName Application -Source 'MsiInstaller' -Newest 20 | Select-Object
TimeGenerated, Message
```

SECTION 6

Troubleshooting Techniques

Diagnosing and resolving GPO application failures

6.1 GPO Not Applying — Diagnostic Checklist

When a GPO setting does not appear to be applied on a target machine, work through this checklist before escalating:

#	Check	Tool / Command
1	Is the machine/user in the correct OU?	Active Directory Users & Computers
2	Is the GPO linked to that OU and link enabled?	GPMC > OU > Linked GPOs tab
3	Does the machine/user have Read & Apply GPO permissions?	GPO > Delegation tab > Advanced
4	Is the user/computer in the Security Filtering group?	GPO > Scope tab > Security Filtering
5	Does a WMI filter block application?	GPO > Scope tab > WMI Filtering
6	Is Block Inheritance set on an intermediate OU?	GPMC > OU > right-click > Properties
7	Is a higher-priority GPO overriding the setting?	gpresult /h report.html
8	Has policy refreshed since the change?	gpupdate /force on target machine
9	Is there a replication delay between DCs?	repadmin /showrepl
10	Are there errors in the Group Policy event log?	Event Viewer > Applications and Services Logs > Microsoft > Windows > GroupPolicy > Operational

6.2 Key Troubleshooting Tools

gpresult — Policy Results Report

The most essential GPO troubleshooting tool. Run on the affected machine:

```
# Summary view — shows applied and denied GPOs
gpresult /r

# HTML report — full detail including all settings
gpresult /h C:\Temp\GPOReport.html
start C:\Temp\GPOReport.html
```

```
# Run as another user (requires admin rights)
gpresult /user domain\username /r

# Verbose output
gpresult /z
```

gpupdate — Force Policy Refresh

```
# Force refresh both Computer and User policy
gpupdate /force

# Force refresh with reboot (for Computer Config changes)
gpupdate /force /boot

# Force refresh remotely (Windows 8+/Server 2012+)
Invoke-GPUpdate -Computer 'PC01' -Force

# Remote refresh via GPMC
# GPMC > OU > right-click > Group Policy Update...
```

RSOP (Resultant Set of Policy)

```
# Launch RSOP wizard (GUI)
rsop.msc

# Note: gpresult /h provides more detail than rsop.msc
# Use RSOP for quick visual confirmation only
```

Group Policy Modeling (GPMC)

Use Group Policy Modeling in GPMC to simulate what policy would apply to a user/computer before actually deploying — invaluable for testing without touching production:

- GPMC > Forest > Domains > [domain] > right-click 'Group Policy Modeling' > Wizard
- Specify the target user and computer. Simulate site, slow link, loopback processing.
- Review the modeled policy result before linking any GPO.

Event Viewer — Group Policy Operational Log

```
# Open Group Policy operational log
eventvwr.msc
# Navigate: Applications and Services Logs > Microsoft > Windows > GroupPolicy >
Operational


# Filter via PowerShell
```

```
Get-WinEvent -LogName 'Microsoft-Windows-GroupPolicy/Operational' -MaxEvents 50 |
  Where-Object { $_.LevelDisplayName -eq 'Error' -or $_.LevelDisplayName -eq
'Warning' } |
  Select-Object TimeCreated, Id, Message | Format-List
```

6.3 Common Issues & Fixes

Symptom	Root Cause	Fix
GPO shows in gpresult but setting not applied	Another GPO with higher priority overrides it	Check link order; use Enforced flag if necessary
GPO listed under 'Denied GPOs'	Missing 'Apply Group Policy' permission	Add user/computer to Security Filtering group
GPO not listed in gpresult at all	Not linked to the OU, or link disabled	Verify GPO link in GPMC; enable the link
Setting applies to admins but not standard users	Policy scoped to Computer Config but processed under User	Verify Computer vs User Config path; check loopback
USB blocking breaks keyboard/mouse	Overly broad block including HID devices	Whitelist keyboard/mouse Hardware IDs
Software not deploying via GPO	MSI on mapped drive, or computer lacks share read rights	Use UNC path; grant Domain Computers Read to share
Folder redirection fails at logon	Network path unavailable at logon time	Ensure DFS/file server is reachable; check DNS
GPO applies in test OU but not production	Block Inheritance on production OU	Check OU properties; use Enforced on GPO if required
DC replication lag causing inconsistent GPO	New GPO not yet replicated to all DCs	Run repadmin /syncall /AdeP; wait for replication

6.4 Event Log Reference

Event ID	Source	Meaning
4016	GroupPolicy	Group Policy processing started for the computer
5016	GroupPolicy	Group Policy processing completed successfully
1085	GroupPolicy	Windows failed to apply Group Policy — contains the failing extension name
1129	GroupPolicy	Processing of Group Policy failed — network connectivity issue to DC
6416	Security	A new external device was recognized — triggered by USB insertion

Event ID	Source	Meaning
4657	Security	Registry value modified — useful for tracking policy setting changes
1000/1001	MsiInstaller	Application install/uninstall event — confirm GPO software deployment success

SECTION 7

GPO Best Practices & Etiquette

The professional standards that separate good GPO administration from great

GPO etiquette refers to the discipline, habits, and professional standards that responsible GPO administrators apply — protecting the environment, the team, and the end-users who depend on stable, predictable policy. These practices are not optional extras; they are the difference between an environment that works reliably and one that generates recurring incidents.

7.1 Naming Conventions

Consistent, descriptive naming is the single most important GPO etiquette rule. A GPO's name should tell any administrator exactly what it does, where it applies, and why — without needing to open it.

NAMING FORMAT: [CATEGORY]-[Description]-[Target] — e.g., SEC-USBBlock-Workstations, APP-AdobeReader-AllUsers, USR-FolderRedirect-Finance

Prefix	Category	Example
SEC-	Security settings	SEC-PasswordPolicy-Domain, SEC-USBBlock-Workstations
APP-	Software deployment	APP-AdobeReader-AllPCs, APP-InternalTool-Finance
USR-	User configuration	USR-FolderRedirect-All, USR-DriveMaps-Finance
COMP-	Computer configuration	COMP-WinUpdate-Servers, COMP-FirewallBaseline-All
DEPT-	Department-specific	DEPT-HR-Printers, DEPT-IT-AdminTools
TEST-	Test/pilot GPOs	TEST-NewSoftware-PilotGroup
DISABLED-	Decommissioned GPOs	DISABLED-OldDriveMap-Finance (keep 30 days then delete)

7.2 Design Principles

One GPO — One Purpose

Create GPOs with a single, clearly defined function. Do not combine USB blocking with software deployment and drive maps in one GPO. Single-purpose GPOs are easier to troubleshoot, modify, and decommission without impacting unrelated settings.

Minimum Scope — Maximum Precision

Link GPOs to the lowest-level OU that contains all intended targets. Never link to a higher OU 'just to be safe' — this risks applying settings to objects that should not receive them.

Avoid Redundant Settings

Do not configure the same setting in multiple GPOs unless deliberately using priority/override logic. Redundant settings create confusion during troubleshooting and make it unclear which GPO is the authoritative source.

Prefer Computer Configuration for Security

Security settings should generally live in Computer Configuration, not User Configuration. Computer-applied policies enforce settings regardless of which user logs in, providing a stronger security guarantee.

> **TIP:** Use Group Policy Modeling (GPMC) before linking any new GPO to predict exactly which settings will apply and whether any existing policies are overriding your new settings.

7.3 Change Management Etiquette

- **Never edit a GPO in production without a change request.** All GPO changes must be raised, reviewed, and approved through the change management process — no matter how minor they appear.
- **Test before you trust.** Every GPO must be tested in a non-production OU or lab environment before linking to production. No exceptions.
- **Never edit a GPO that another team member is working on.** GPOs do not have edit locking. Communicate with the team before editing shared or complex GPOs to avoid overwriting each other's changes.
- **Use comments in the GPO.** In GPMC, every GPO has a Comment field (GPO Properties > Comment). Always enter a brief description of the GPO's purpose, the date it was created, and the change request number.
- **Don't use Default Domain Policy as a catch-all.** Default Domain Policy should contain only domain-wide settings it was designed for (password policy, account lockout). Do not add custom settings here — create dedicated GPOs instead.
- **Disable unused halves of GPOs.** If a GPO only uses Computer Configuration settings, disable the User Configuration node (and vice versa). This reduces processing overhead and improves client logon/startup times.

> **NOTE:** To disable unused configuration: In GPMC, right-click the GPO > GPO Status > select 'User Configuration settings disabled' or 'Computer Configuration settings disabled'.

7.4 Documentation Standards

Every GPO in production must have a corresponding entry in the GPO Register — a centrally maintained document or ITSM record that captures:

Field	Required Content
GPO Name	Exact name as it appears in GPMC
GUID	GPO GUID for audit/troubleshooting reference
Purpose	Plain-language description of what the GPO does and why
Linked OUs	All OUs to which this GPO is linked
Target Scope	Security group(s) in Security Filtering
Created Date	Date of initial creation
Created By	Admin who created the GPO
Change Request	Associated CR or ticket number
Last Modified	Date and nature of most recent change
Review Date	Scheduled annual review date

7.5 Security & Delegation

- **Limit GPO Edit rights strictly.** Only designated GPO administrators should have the ability to edit GPOs. Use the Delegation tab in GPMC to control who has Read, Edit, Edit/Delete/Modify Security permissions.
- **Audit GPO changes.** Enable Advanced Audit Policy for 'Audit Policy Change' and 'Audit Directory Service Changes' so all GPO modifications are captured in the Security event log.
- **Review GPOs annually.** Schedule an annual review of all production GPOs. Decommission GPOs for retired applications, departed departments, or superseded settings.
- **Protect Default Domain Policy and Default Domain Controllers Policy.** These two GPOs are critical to domain function. Apply 'Enforced' flag only with extreme care. Never delegate edit rights to junior admins.
- **Back up GPOs before changes.** Use GPMC Backup (right-click GPO > Back Up) before making any significant change. Store backups in a versioned, accessible location.

```
# Backup all GPOs in the domain via PowerShell
Backup-GPO -All -Path C:\GPOBackups\$(Get-Date -f 'yyyy-MM-dd')
```

```
# Restore a specific GPO from backup
Restore-GPO -Name 'SEC-USBBlock-Workstations' -Path C:\GPOBackups\2025-01-15


# List all GPOs with creation and modification dates
Get-GPO -All | Select-Object DisplayName, CreationTime, ModificationTime | Sort-
Object CreationTime
```

SECTION 8

Quick Reference

Commands, checklists, and at-a-glance tables

8.1 Essential GPO Commands

Command	Purpose
gpupdate /force	Force immediate policy refresh on current machine
gpresult /r	Show applied GPOs summary for current user/computer
gpresult /h report.html	Generate full HTML GPO report
rsop.msc	Open Resultant Set of Policy GUI
gpmc.msc	Open Group Policy Management Console
Invoke-GPUpdate -Computer PC01 -Force	Force remote policy refresh via PowerShell
Get-GPO -All	List all GPOs in the domain
Backup-GPO -All -Path C:\Backup	Backup all domain GPOs
repadmin /showrepl	Check DC replication status
repadmin /syncall /AdeP	Force AD replication across all DCs

8.2 GPO Deployment Pre-Flight Checklist

- GPO has a clear, convention-compliant name
- GPO purpose documented in the Comment field
- Change request raised and approved
- Settings tested in lab/test OU
- Security Filtering configured with target group (Authenticated Users removed if scoping to specific group)
- WMI Filter set if OS/hardware conditions apply
- Unused configuration node (User or Computer) disabled
- GPO backed up before any change
- Link order verified relative to other GPOs on the OU
- GPO Register updated with all required fields
- Post-deployment validation completed via gpresult

8.3 USB Blocking — Policy Path Summary

Setting	Full Policy Path
Block all USB storage	Computer Config > Admin Templates > System > Removable Storage Access > All Removable Storage: Deny all access
Disable USB write	Computer Config > Admin Templates > System > Removable Storage Access > Removable Disks: Deny write access
Prevent device install	Computer Config > Admin Templates > System > Device Installation > Device Installation Restrictions > Prevent installation of removable devices
Whitelist device by ID	Computer Config > Admin Templates > System > Device Installation > Device Installation Restrictions > Allow installation of devices matching these device IDs

8.4 Folder Redirection — Policy Path Summary

Folder	Policy Path
Documents	User Config > Policies > Windows Settings > Folder Redirection > Documents
Desktop	User Config > Policies > Windows Settings > Folder Redirection > Desktop
Downloads	User Config > Policies > Windows Settings > Folder Redirection > Downloads
AppData	User Config > Policies > Windows Settings > Folder Redirection > AppData (Roaming)

8.5 MSI Deployment — Policy Path

Deployment Type	Policy Path
Computer-assigned (silent, all users)	Computer Config > Policies > Software Settings > Software Installation
User-assigned (at logon)	User Config > Policies > Software Settings > Software Installation
User-published (optional install)	User Config > Policies > Software Settings > Software Installation (Published type)